

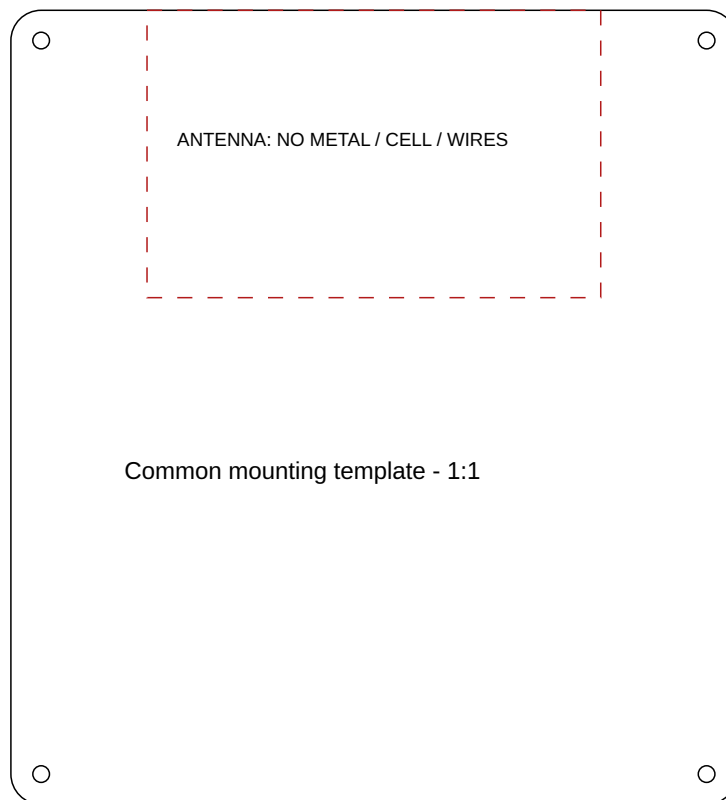
Keychain Kreatures - flat, removable board stack

C.6 / P.3 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

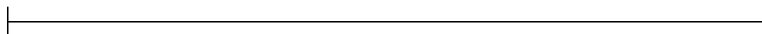
Shared outline: 96 x 105 mm; 4 mm corner radius; 1.6 mm PCB.

Four 2.2 mm clearance holes: (4,4), (92,4), (4,101), (92,101) mm.

Coordinates below: top-left origin, X right, Y down. Both boards use this pattern.



100 mm calibration line:



PRINT AT 100% / ACTUAL SIZE. Disable fit-to-page. Measure the line first.

Stack orientation: main display/buttons outward at the front of the case.

Power component side faces the rear cover; its test-pad face points toward main.

Power-board front view is X-mirrored relative to main-front physical coordinates.

Start a fit mock-up with FOUR 20 mm M2 insulating standoffs between PCB faces.

This spacing is provisional: check plugged JST bodies, cable bends and solder tails.

Clip solder tails to ≤ 2.5 mm, inspect every joint, and keep loose wires restrained.

Unscrew and unplug the rear power board to expose the student main-board parts.

Do not trap a battery between solder tails and components. Cell position is not frozen.

Keep a protective insulating barrier; never use battery foil as an electrical insulator.

Flat parts and assembly limits

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R1-R43

Existing Yageo MFR-25 values

All horizontal, 10.16 mm lead pitch

C1,C3,C5,C13,C17,C25

Panasonic ECEA1CKA101

100 uF / 16 V; max flat height 7.3 mm*

C9,C11,C15,C21

Panasonic ECEA1CKA100

10 uF / 16 V; max flat height 5.0 mm*

Q1-Q6

Existing exact transistor MPNs

Flat marked face up; preserve pin order

* Manufacturer maximum can diameter plus 0.5 mm insulating support; not scaled artwork.

Both capacitor types use the full 8 mm maximum body-length courtyard.

Capacitor square pad 1 is POSITIVE. Sleeve stripe marks NEGATIVE pad 2.

10 uF native lead pitch is 1.5 mm: form to the 2.54 mm PCB pitch without crossing.

Support leads at the seal while bending; do not pull, twist, or stress the rubber seal.

Use nonconductive support under horizontal cans. Do not obstruct the pressure vent.

TO-92 bodies extend toward the footprint body rectangle; marked flat face stays up.

Q6 is LP0701N3-G, not BC327. Verify actual pinout before forming or soldering.

IR lenses must still face the case opening; optical parts are not flattened blindly.

Sockets, screen, buttons and the front RGB retain their actual functional height.

The socketed ESP32 rear envelope is about 15.3 mm; it, not the caps, limits spacing.

The socketed RGB front envelope remains about 22.1 mm. Case depth is NOT final.

Power board remains a factory-assembled SMT subassembly, not a student SMT kit.

Prototype checks still required

Measure delivered screen, ESP32, SD module, sockets, plugs and standoffs.

Check horizontal capacitor ripple/rail transients during Wi-Fi, audio and motor bursts.

Check component temperatures, charging, rail sequencing and battery protection.

Confirm antenna performance with the actual rear board, battery, harness and case.

Prototype files issued separately; approve physical fit and factory DFM before ordering.

Main-board rear - populated mechanical preview

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Preview is not a powered test. Seller-module envelopes and formed leads need sample checks.

Use C6_flat_stack and P3_matching_stack, not the old manufacturing ZIPs.

Power-board rear-cover side - matching outline

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Routed power core preserved. Empty upper antenna region must stay clear of metal.

Power J3 -> main J1: 1=5V, 2=GND, 3=3.3V, 4=3.2V. Keyed pin-to-pin harness.

This is the power FRONT view; mirror X to compare with the main FRONT in the stack.